

Qualification Pack



Solar Photovoltaic Site Survey Assistant

QP Code: SGJ/Q1203

Version: 1.0

NSQF Level: 3.5

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SGJ/Q1203: Solar Photovoltaic Site Survey Assistant

Brief Job Description

Solar Photovoltaic Site Survey Assistant performs basic solar site survey and assessment for the installation of small solar system. He/She conducts a physical solar site survey using various tools and ensures compliance with state and national regulations, guidelines, along with industry and safety standards for solar PV installations. The job holder must show a high level of attention to while surveying and have good written and verbal skills to communicate with supervisor, solar consumers and fellow team members.

Personal Attributes

This job requires the individual to concentrate on the job at hand and complete it without any accidents so diligence and hardworking are desired attributes for individuals performing this role. He/She must also demonstrate strong work ethics, an ability to communicate courteously with co-workers and must be good with following instructions of the supervisor.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [SGJ/N1209: Introduce Solar Energy applications and importance of site survey](#)
2. [SGJ/N1210: Introduce Solar Photovoltaic Plant](#)
3. [SGJ/N1211: Assess shadow free area at solar project site](#)
4. [SGJ/N1212: Perform the use of tools for site survey](#)
5. [SGJ/N1213: Perform load analysis for small solar consumers](#)
6. [SGJ/N1214: Assess Consumers need for solar power](#)
7. [SGJ/N1215: Prepare a site survey report](#)
8. [SGJ/N1216: Maintain Personal Health & Safety while performing site survey at solar project site](#)
9. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Green Jobs
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Sub-Sector	Renewable Energy
Occupation	Site survey
Country	India
NSQF Level	3.5
Credits	13
Aligned to NCO/ISCO/ISIC Code	NCO-2015/7422.1801
Minimum Educational Qualification & Experience	10th Class (pursuing continuous schooling) with NA of experience OR 11th grade pass (Or Completed 1st year of 3 year diploma after 10th) with NA of experience OR 8th grade pass with 2 year NTC plus 1 year NAC plus 1 year CITS OR 8th grade pass with 3 Years of experience OR Previous relevant Qualification of NSQF Level (3 Experience) with 1-2 Years of experience OR Previous relevant Qualification of NSQF Level (2.5) with 3 Years of experience
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	16 Years
Last Reviewed On	NA
Next Review Date	31/08/2026
NSQC Approval Date	31/08/2023
Version	1.0
Reference code on NQR	QG-3.5-ES-00909-2023-V1-SCGJ
NQR Version	1

Remarks:



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Total 360 Hours i.e. 12 Credits (330 Hours of Mandatory NOS theory & practical and 30 hours of Employability Skills)

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SGJ/N1209: Introduce Solar Energy applications and importance of site survey

Description

This unit is about introduction to Solar Energy and importance of site survey for various solar applications

Scope

The scope covers the following :

- solar energy and its use in various applications
- irradiation and its significance in power generation from solar Photovoltaic

Elements and Performance Criteria

Introduction to Solar Energy and importance of site survey for various solar applications

To be competent, the user/individual on the job must be able to:

- PC1.** discuss the basics of solar energy/ electricity and electrical concepts and perform simple calculations to illustrate the fundamental concepts of power and energy
- PC2.** discuss the role of Assistant Solar Site Surveyor and emerging jobs & entrepreneurial opportunities
- PC3.** discuss the advantages of doing this course.
- PC4.** discuss the movement of sun and sun path diagram and assess its effect on the performance of the solar energy generation and demonstrate how the movement of sun affects the solar energy generation.
- PC5.** show how to interpret signs, notices and/or cautions at project site

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Company's Surveying Policy
- KU2.** legal formalities applicable for the particular business
- KU3.** knowledge of processes involved in the business
- KU4.** the basics of solar energy/ electricity and electrical concepts
- KU5.** the role of Assistant Solar Site Surveyor and emerging jobs & entrepreneurial opportunities
- KU6.** the advantages of doing this course
- KU7.** the movement of sun and sun path diagram and assess its effect on the performance of the solar energy generation
- KU8.** show the fundamental concepts of power and energy
- KU9.** show the movement of sun affects the solar energy generation
- KU10.** show how to interpret signs, notices and/or cautions at project site

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Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** complete documentation and work applicable to the role
- GS2.** read and write in vernacular/English language
- GS3.** read various color codes, as per standard electrical , maintenance and civil nomenclature
- GS4.** express statements with required clarity
- GS5.** participate and understand the main points of simple discussions
- GS6.** follow organization's rule-based decision making process
- GS7.** work constructively and collaboratively with others
- GS8.** follow organization's code of conduct
- GS9.** approach relevant authority when required
- GS10.** ask question for better understanding
- GS11.** communicating with team members and colleagues on the significance of greening of jobs
- GS12.** complete documentation work applicable to the role

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Solar Energy and importance of site survey for various solar applications</i>	30	20	-	-
PC1. discuss the basics of solar energy/ electricity and electrical concepts and perform simple calculations to illustrate the fundamental concepts of power and energy	7	7	-	-
PC2. discuss the role of Assistant Solar Site Surveyor and emerging jobs & entrepreneurial opportunities	8	-	-	-
PC3. discuss the advantages of doing this course.	8	-	-	-
PC4. discuss the movement of sun and sun path diagram and assess its effect on the performance of the solar energy generation and demonstrate how the movement of sun affects the solar energy generation.	7	5	-	-
PC5. show how to interpret signs, notices and/or cautions at project site	-	8	-	-
NOS Total	30	20	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N1209
NOS Name	Introduce Solar Energy applications and importance of site survey
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Site survey
NSQF Level	3.5
Credits	1
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	31/08/2026
NSQC Clearance Date	31/08/2023

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SGJ/N1210: Introduce Solar Photovoltaic Plant

Description

This unit is about basics of Solar PV Plant

Scope

The scope covers the following :

- Basics of Solar Photovoltaic Plant technology
- type of solar cell and solar module

Elements and Performance Criteria

Basics of Solar PV Plant

To be competent, the user/individual on the job must be able to:

- PC1.** discuss the basics of Solar PV plant and its working
- PC2.** discuss the key components of Solar PV plant and show how to identify the key components of solar Photovoltaic plant and demonstrate their functions
- PC3.** discuss the basics of grid connected, off grid and hybrid solar PV system and draw a schematic of a grid connected, off grid and hybrid solar Photovoltaic plant
- PC4.** discuss the advantages of grid connected, off grid and hybrid solar PV system
- PC5.** discuss basics of Solar module operation and array formation
- PC6.** discuss the standard test condition of solar PV module
- PC7.** discuss the basics of working of Solar Inverter and Its type
- PC8.** discuss the operation of Solar PV plant and show the basic operation of a solar power system
- PC9.** discuss the basics of applications such as Solar Pump and solar street light
- PC10.** show how Solar power plant produce electricity
- PC11.** show the relevance of grid connected, off grid and hybrid solar PV system

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** policies and procedures that impact the business process
- KU2.** local corporations / authorities that have a bearing on the particular business process
- KU3.** reporting procedure in case of deviations
- KU4.** knowledge of processes involved in the business
- KU5.** the importance of conducting site survey for various solar projects
- KU6.** the basics of site survey
- KU7.** the azimuth angle and its importance
- KU8.** assess the available shadow free area at various project site

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- KU9.** using the compass at site
- KU10.** assess the correct orientation and tilt angle
- KU11.** to identify shadow casting object
- KU12.** the basics of various software (e.g. google earth) and tools used for performing site survey
- KU13.** show shadow casting object at site
- KU14.** show how shadow impacts the solar generation and could damage solar panel by creating hot spot
- KU15.** show the basics use of site survey through various software (e.g. google earth) and other tools

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** complete documentation work applicable to the role
- GS2.** read and write in vernacular/English language
- GS3.** read various color codes, as per standard electrical, mechanical and civil nomenclature
- GS4.** express statements with required clarity
- GS5.** participate in and understand the main points of simple discussions
- GS6.** follow organization's rule-based decision making process
- GS7.** work constructively and collaboratively with others
- GS8.** follow organization's code of conduct
- GS9.** approach relevant authority when required
- GS10.** ask questions for better understanding
- GS11.** communicate with team members and colleagues on the significance of greening of jobs
- GS12.** complete documentation work applicable to the role

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Basics of Solar PV Plant</i>	25	25	-	-
PC1. discuss the basics of Solar PV plant and its working	2	-	-	-
PC2. discuss the key components of Solar PV plant and show how to identify the key components of solar Photovoltaic plant and demonstrate their functions	2	5	-	-
PC3. discuss the basics of grid connected, off grid and hybrid solar PV system and draw a schematic of a grid connected, off grid and hybrid solar Photovoltaic plant	3	5	-	-
PC4. discuss the advantages of grid connected, off grid and hybrid solar PV system	3	-	-	-
PC5. discuss basics of Solar module operation and array formation	3	-	-	-
PC6. discuss the standard test condition of solar PV module	3	-	-	-
PC7. discuss the basics of working of Solar Inverter and Its type	3	-	-	-
PC8. discuss the operation of Solar PV plant and show the basic operation of a solar power system	3	5	-	-
PC9. discuss the basics of applications such as Solar Pump and solar street light	3	-	-	-
PC10. show how Solar power plant produce electricity	-	5	-	-
PC11. show the relevance of grid connected, off grid and hybrid solar PV system	-	5	-	-
NOS Total	25	25	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N1210
NOS Name	Introduce Solar Photovoltaic Plant
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Site survey
NSQF Level	3.5
Credits	1
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	31/08/2026
NSQC Clearance Date	31/08/2023

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SGJ/N1211: Assess shadow free area at solar project site

Description

This unit is about assess shadow free area at solar project site

Scope

The scope covers the following :

- Introduction to Site survey
- Importance of assessing shadow free area

Elements and Performance Criteria

Assess shadow free area at solar project site

To be competent, the user/individual on the job must be able to:

- PC1.** discuss the importance of conducting site survey for various solar projects (e.g. mini grid, grid connected rooftop etc
- PC2.** explain basics of site survey
- PC3.** discuss the azimuth angle and its importance
- PC4.** discuss how to assess the available shadow free area at various project site and demonstrate how shadow impacts the solar generation and could damage solar panel by creating hot spot
- PC5.** discuss how to use the compass at site
- PC6.** discuss to assess the correct orientation and tilt angle
- PC7.** explain how to identify shadow casting object and show how to identify shadow casting object at site
- PC8.** discuss the basics of various software (e.g. google earth) and tools used for performing site survey and demonstrate the basics use of site survey through various software (e.g google earth) and other tools

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** local corporations / authorities that have a bearing on the particular business process
- KU2.** policies and procedures that impact the business process
- KU3.** legal formalities applicable for the particular business
- KU4.** the importance of having an ethical and value based approach to governance
- KU5.** Company's surveying Policy
- KU6.** Company's documentation policy
- KU7.** Recognize problems and search for solutions.
- KU8.** Choose best methods to complete assigned tasks.

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- KU9.** Apply domain knowledge, observations and data to select course of action to perform tasks related to Solar Photovoltaic Systems.
- KU10.** Manage relationships with customers with intent on satisfying its requirements for service delivery.
- KU11.** Planning and organization of work to meet deadlines.
- KU12.** Work constructively and collaboratively with others

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** prepare legal formats
- GS2.** do's and don'ts of material handling and storage.
- GS3.** read and understand statutory documents
- GS4.** Read English and/or vernacular language.
- GS5.** Read and understand manuals, health and safety instructions, memos, other company documents.
- GS6.** Ability to read from different sources- books screens in machines and signage.
- GS7.** Understand the various color codes, as per standard electrical, mechanical
- GS8.** Express statements or information clearly so that others can hear and understand.
- GS9.** Participate in and understand the main points of simple discussions.
- GS10.** Respond appropriately to any queries.
- GS11.** Follow organization rule-based decision making process.
- GS12.** Planning and organization of work to meet deadlines.
- GS13.** Follow code of conduct.
- GS14.** Ask questions for better understanding
- GS15.** Critically evaluate information obtained from customers, supervisor and coworkers to perform day to day activities.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Assess shadow free area at solar project site</i>	35	15	-	-
PC1. discuss the importance of conducting site survey for various solar projects (e.g. mini grid, grid connected rooftop etc)	5	-	-	-
PC2. explain basics of site survey	5	-	-	-
PC3. discuss the azimuth angle and its importance	5	-	-	-
PC4. discuss how to assess the available shadow free area at various project site and demonstrate how shadow impacts the solar generation and could damage solar panel by creating hot spot	4	5	-	-
PC5. discuss how to use the compass at site	4	-	-	-
PC6. discuss to assess the correct orientation and tilt angle	4	-	-	-
PC7. explain how to identify shadow casting object and show how to identify shadow casting object at site	4	5	-	-
PC8. discuss the basics of various software (e.g. google earth) and tools used for performing site survey and demonstrate the basics use of site survey through various software (e.g google earth) and other tools	4	5	-	-
NOS Total	35	15	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N1211
NOS Name	Assess shadow free area at solar project site
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Site survey
NSQF Level	3.5
Credits	1
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	31/08/2026
NSQC Clearance Date	31/08/2023

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SGJ/N1212: Perform the use of tools for site survey

Description

This unit is about perform the use of tools for site survey

Scope

The scope covers the following :

- how to use software and tools to perform Site survey
- Importance of shadow free area

Elements and Performance Criteria

Perform the use of tools for site survey

To be competent, the user/individual on the job must be able to:

- PC1.** discuss how to perform site survey using various tools and show how to perform site survey at various project sites using various tools
- PC2.** discuss the tools used for measuring the current and voltage
- PC3.** discuss the tools used to find the correct orientation and azimuth angle and show how to perform the use of tools for assessing correct orientation and azimuth angle
- PC4.** discuss the use of measuring tape and compass and show how to use measuring tape and compass
- PC5.** discuss the use of google earth for site survey and show the basic use of software tools like google earth for site survey
- PC6.** discuss the concept and basic use of other software tools/apps for conducting shadow analysis and estimating PV capacity and show how to perform basic use of other software tools/Apps for conducting shadow analysis and estimating Photovoltaic capacity
- PC7.** show how to measure various parameters like current and voltage

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Planning and organization of work to meet deadlines.
- KU2.** Follow organization rule-based decision making process.
- KU3.** Follow code of conduct.
- KU4.** make timely decisions for efficient utilization of resources
- KU5.** Apply domain knowledge, observations and data to select course of action to perform tasks related to Solar Photovoltaic Systems.
- KU6.** Manage relationships with customers with intent on satisfying its requirements for service delivery.
- KU7.** to perform site survey using various tools
- KU8.** tools used for measuring the current and voltage

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- KU9.** tools used to find the correct orientation and azimuth angle
- KU10.** use of measuring tape and compass
- KU11.** use of google earth for site survey
- KU12.** use of other software tools/apps for conducting shadow analysis and estimating PV capacity
- KU13.** site survey at various project sites using various tools
- KU14.** use of tools for assessing correct orientation and azimuth angle
- KU15.** use of measuring tape and compass
- KU16.** measure various parameters like current and voltage
- KU17.** use of software tools like google earth for site survey
- KU18.** conducting shadow analysis and estimating PV capacity

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Read English and/or vernacular language.
- GS2.** Fill up documentation applicable to one's role.
- GS3.** read and understand statutory documents
- GS4.** communicate with concerned officials to discuss statutory and legal requirements
- GS5.** make appropriate decisions regarding responsibilities of job roles
- GS6.** Express statements or information clearly so that others can hear and understand.
- GS7.** Respond appropriately to any queries.
- GS8.** Follow organization rule-based decision making process.
- GS9.** Recognize problems and search for solutions.
- GS10.** Choose best methods to complete assigned tasks
- GS11.** Approach relevant authority when required.
- GS12.** Critically evaluate information obtained from customers, supervisor and coworkers to perform day to day activities
- GS13.** Ask questions for better understanding.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Perform the use of tools for site survey</i>	25	25	-	-
PC1. discuss how to perform site survey using various tools and show how to perform site survey at various project sites using various tools	4	5	-	-
PC2. discuss the tools used for measuring the current and voltage	4	-	-	-
PC3. discuss the tools used to find the correct orientation and azimuth angle and show how to perform the use of tools for assessing correct orientation and azimuth angle	4	3	-	-
PC4. discuss the use of measuring tape and compass and show how to use measuring tape and compass	4	3	-	-
PC5. discuss the use of google earth for site survey and show the basic use of software tools like google earth for site survey	4	5	-	-
PC6. discuss the concept and basic use of other software tools/apps for conducting shadow analysis and estimating PV capacity and show how to perform basic use of other software tools/Apps for conducting shadow analysis and estimating Photovoltaic capacity	5	4	-	-
PC7. show how to measure various parameters like current and voltage	-	5	-	-
NOS Total	25	25	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N1212
NOS Name	Perform the use of tools for site survey
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Site survey
NSQF Level	3.5
Credits	2
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	31/08/2026
NSQC Clearance Date	31/08/2023

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SGJ/N1213: Perform load analysis for small solar consumers

Description

This unit is about perform load analysis for small solar consumers

Scope

The scope covers the following :

- perform load analysis
- types of load and how to prepare load profile
- how to interpret electricity bills

Elements and Performance Criteria

Perform load analysis for small solar consumers

To be competent, the user/individual on the job must be able to:

- PC1.** discuss how to perform basic load analysis of a small solar consumer
- PC2.** discuss the basics types of loads and its characteristics and demonstrate how to identify various load type
- PC3.** discuss monthly electricity bill and various parameters mentioned on the bill
- PC4.** discuss the basics of contracted load, connected load and running load and its impact of solar PV system selection and show how to estimate and perform calculation for contracted load, connected load etc.
- PC5.** show how to identify various load type
- PC6.** show the characteristics and variations in load demand (e.g. seasonal)
- PC7.** show how to do the basic analysis of an electricity bill of a small consumer

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Company's work safety policy
- KU2.** Company's documentation policy.
- KU3.** Company's different department and concerned authority.
- KU4.** Company's reporting structure and Organization culture
- KU5.** basic load analysis of a small solar consumer
- KU6.** basics types of loads and its characteristics
- KU7.** monthly electricity bill and various parameters mentioned on the bill
- KU8.** basics of contracted load, connected load and running load and its impact of solar PV system selection
- KU9.** the load to be connected to the Solar PV system and how to prepare the load profile
- KU10.** the load to be connected to the Solar PV system and how to prepare the load profile

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KU11. the characteristics and variations in load demand (e.g. seasonal)

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** complete documentation work applicable to the role
- GS2.** read and write in vernacular/English language
- GS3.** read various color codes, as per standard electrical, mechanical and civil nomenclature
- GS4.** do the calculations accurately
- GS5.** express statements with required clarity
- GS6.** participate in and understand the main points of simple discussions
- GS7.** follow organization's rule-based decision making process
- GS8.** work constructively and collaboratively with others
- GS9.** follow organization's code of conduct
- GS10.** approach relevant authority when required
- GS11.** ask questions for better understanding
- GS12.** communicate with team members and colleagues on the significance of greening of jobs
- GS13.** complete documentation work applicable to the role

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Perform load analysis for small solar consumers</i>	25	25	-	-
PC1. discuss how to perform basic load analysis of a small solar consumer	6	-	-	-
PC2. discuss the basics types of loads and its characteristics and demonstrate how to identify various load type	6	5	-	-
PC3. discuss monthly electricity bill and various parameters mentioned on the bill	7	-	-	-
PC4. discuss the basics of contracted load, connected load and running load and its impact of solar PV system selection and show how to estimate and perform calculation for contracted load, connected load etc.	6	5	-	-
PC5. show how to identify various load type	-	5	-	-
PC6. show the characteristics and variations in load demand (e.g. seasonal)	-	5	-	-
PC7. show how to do the basic analysis of an electricity bill of a small consumer	-	5	-	-
NOS Total	25	25	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N1213
NOS Name	Perform load analysis for small solar consumers
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Site survey
NSQF Level	3.5
Credits	2
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	31/08/2026
NSQC Clearance Date	31/08/2023

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SGJ/N1214: Assess Consumers need for solar power

Description

This unit is about assess Consumers need for solar power

Scope

The scope covers the following :

- how to assess consumer's need
- requirement of the consumer
- energy demand and load connected

Elements and Performance Criteria

Assess Consumer's need for solar power

To be competent, the user/individual on the job must be able to:

- PC1.** discuss how to interact with concerned customers to identify their requirement and demonstrate how to interact with the customer
- PC2.** discuss how to meet the requirements as per consumer's budget
- PC3.** discuss how to identify consumer's concern, if any
- PC4.** discuss the scope of work of consumers
- PC5.** show the load to be connected to the solar PV system

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Company's work safety policy
- KU2.** Company's documentation policy
- KU3.** Obtain authorization from specified field safety officer and supervisor.
- KU4.** Company's different department and concerned authority
- KU5.** Company's reporting structure and Organization culture
- KU6.** concerned customers to identify their requirement
- KU7.** the requirements as per consumer's budget
- KU8.** the consumer's concern, if any
- KU9.** interaction with the customer
- KU10.** the load to be connected to the Solar PV system

Generic Skills (GS)

User/individual on the job needs to know how to:

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- GS1.** complete documentation work applicable to the role
- GS2.** read and write in vernacular/English language
- GS3.** read various color codes, as per standard electrical, mechanical and civil nomenclature
- GS4.** express statements with required clarity
- GS5.** participate in and understand the main points of simple discussions
- GS6.** follow organization's rule-based decision making process
- GS7.** work constructively and collaboratively with others
- GS8.** follow organization's code of conduct
- GS9.** approach relevant authority when required
- GS10.** ask questions for better understanding
- GS11.** communicate with team members and colleagues on the significance of greening of jobs
- GS12.** complete documentation work applicable to the role

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Assess Consumer's need for solar power</i>	33	17	-	-
PC1. discuss how to interact with concerned customers to identify their requirement and demonstrate how to interact with the customer	8	8	-	-
PC2. discuss how to meet the requirements as per consumer's budget	8	-	-	-
PC3. discuss how to identify consumer's concern, if any	8	-	-	-
PC4. discuss the scope of work of consumers	9	-	-	-
PC5. show the load to be connected to the solar PV system	-	9	-	-
NOS Total	33	17	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N1214
NOS Name	Assess Consumers need for solar power
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Site survey
NSQF Level	3.5
Credits	1
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	31/08/2026
NSQC Clearance Date	31/08/2023

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SGJ/N1215: Prepare a site survey report

Description

This unit is about prepare a site survey report

Scope

The scope covers the following :

- Introduction of Site survey report
- how to prepare a site survey report
- Identify the major obstacle on site

Elements and Performance Criteria

Prepare a site survey report

To be competent, the user/individual on the job must be able to:

- PC1.** discuss the key part of a solar site survey report and show how to identify site parameters and discuss the importance of performing site survey for solar projects
- PC2.** discuss what information are required and identify how to get those for analysis and to be included into the report
- PC3.** discuss how to Assess the location, any site level prerequisites and optimise the route plan and show how to prepare a site map of the location where installation has to be carried out
- PC4.** discuss how to prepare a site map and show how to perform the site survey, analyse site parameters and prepare the site survey report
- PC5.** discuss first-hand layout sketch of site and include into the report
- PC6.** discuss the major obstacle on site
- PC7.** discuss technical specification of existing electrical system (Breaker capacity, spare feeder detail, etc)

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Company's Surveying Policy.
- KU2.** Company's work safety policy
- KU3.** Company's documentation policy.
- KU4.** Obtain authorization from specified field safety officer and supervisor.
- KU5.** Company's reporting structure and Organization culture.
- KU6.** Company's different department and concerned authority.
- KU7.** complete tasks efficiently and accurately within stipulated time
- KU8.** the importance of having an ethical and value based approach to governance
- KU9.** Relevant standards and regulations for Surveying of Solar Photovoltaic Power Plant in India

Qualification Pack

KU10. Know how of tools & tackles required to carry out the work

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** complete documentation work applicable to the role
- GS2.** read and write in vernacular/English language
- GS3.** read various color codes, as per standard electrical, mechanical and civil nomenclature
- GS4.** express statements with required clarity
- GS5.** participate in and understand the main points of simple discussions
- GS6.** follow organization's rule-based decision making process
- GS7.** work constructively and collaboratively with others
- GS8.** follow organization's code of conduct
- GS9.** approach relevant authority when required
- GS10.** ask questions for better understanding
- GS11.** communicate with team members and colleagues on the significance of greening of jobs
- GS12.** complete documentation work applicable to the role

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare a site survey report</i>	35	15	-	-
PC1. discuss the key part of a solar site survey report and show how to identify site parameters and discuss the importance of performing site survey for solar projects	3	5	-	-
PC2. discuss what information are required and identify how to get those for analysis and to be included into the report	6	-	-	-
PC3. discuss how to Assess the location, any site level prerequisites and optimise the route plan and show how to prepare a site map of the location where installation has to be carried out	5	5	-	-
PC4. discuss how to prepare a site map and show how to perform the site survey, analyse site parameters and prepare the site survey report	5	5	-	-
PC5. discuss first-hand layout sketch of site and include into the report	6	-	-	-
PC6. discuss the major obstacle on site	5	-	-	-
PC7. discuss technical specification of existing electrical system (Breaker capacity, spare feeder detail, etc)	5	-	-	-
NOS Total	35	15	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N1215
NOS Name	Prepare a site survey report
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Site survey
NSQF Level	3.5
Credits	2
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	31/08/2026
NSQC Clearance Date	31/08/2023

Qualification Pack

SGJ/N1216: Maintain Personal Health & Safety while performing site survey at solar project site

Description

This unit is about maintain Personal Health & Safety while performing site survey at solar project site

Scope

The scope covers the following :

- Requirements for safe work area
- Importance of Occupational health & Safety standards and regulations for performing site survey

Elements and Performance Criteria

Maintain Personal Health & Safety while performing site survey at solar project site

To be competent, the user/individual on the job must be able to:

- PC1.** discuss the requirements for safe work area and demonstrate how to keep the work area safe
- PC2.** discuss how to perform health and safety measures during solar site survey and show what measures have to be taken to ensure safe practices for solar site survey
- PC3.** discuss the importance of administering first aid and demonstrate how to administer first aid.
- PC4.** discuss the personal protective equipment used for the specific purpose and demonstrate the usage of personal protective equipment and tools for ensuring safety during site survey for solar projects.
- PC5.** discuss the hazards associated with solar products assembly, operations, and repair
- PC6.** discuss work safety procedures to be performed at solar site survey
- PC7.** discuss the importance of Occupational health & Safety standards and regulations for performing site survey
- PC8.** discuss good housekeeping practices and infection control guidelines and demonstrate good housekeeping and infection control & prevention practices.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Company's documentation policy
- KU2.** Company's work safety policy
- KU3.** Relevant Personal protective equipment's required for Surveying
- KU4.** the requirements for safe work area
- KU5.** safety measures during solar site survey
- KU6.** the importance of administering first aid
- KU7.** the personal protective equipment used for the specific purpose
- KU8.** the hazards associated with solar products assembly, operations, and repair

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- KU9.** work safety procedures to be performed at solar site survey
- KU10.** the importance of Occupational health & Safety standards and regulations for performing site survey
- KU11.** good housekeeping practices and infection control guidelines
- KU12.** to keep the work area safe
- KU13.** measures to be taken to ensure safe practices for solar site survey
- KU14.** how to administer first aid
- KU15.** the usage of personal protective equipment and tools for ensuring safety during site survey for solar projects
- KU16.** good housekeeping and infection control & prevention practices

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** complete documentation work applicable to the role
- GS2.** read and write in vernacular/English language
- GS3.** read various color codes, as per standard electrical, mechanical and civil nomenclature
- GS4.** express statements with required clarity
- GS5.** participate in and understand the main points of simple discussions
- GS6.** follow organization's rule-based decision making process
- GS7.** work constructively and collaboratively with others
- GS8.** follow organization's code of conduct
- GS9.** approach relevant authority when required
- GS10.** ask questions for better understanding
- GS11.** communicate with team members and colleagues on the significance of greening of jobs
- GS12.** complete documentation work applicable to the role

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Maintain Personal Health & Safety while performing site survey at solar project site</i>	27	23	-	-
PC1. discuss the requirements for safe work area and demonstrate how to keep the work area safe	3	3	-	-
PC2. discuss how to perform health and safety measures during solar site survey and show what measures have to be taken to ensure safe practices for solar site survey	3	5	-	-
PC3. discuss the importance of administering first aid and demonstrate how to administer first aid.	4	5	-	-
PC4. discuss the personal protective equipment used for the specific purpose and demonstrate the usage of personal protective equipment and tools for ensuring safety during site survey for solar projects.	4	5	-	-
PC5. discuss the hazards associated with solar products assembly, operations, and repair	3	-	-	-
PC6. discuss work safety procedures to be performed at solar site survey	3	-	-	-
PC7. discuss the importance of Occupational health & Safety standards and regulations for performing site survey	3	-	-	-
PC8. discuss good housekeeping practices and infection control guidelines and demonstrate good housekeeping and infection control & prevention practices.	4	5	-	-
NOS Total	27	23	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N1216
NOS Name	Maintain Personal Health & Safety while performing site survey at solar project site
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Site survey
NSQF Level	3.5
Credits	1
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	31/08/2026
NSQC Clearance Date	31/08/2023

Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

Qualification Pack

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

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- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	18/02/2026
Next Review Date	18/02/2029
NSQC Clearance Date	18/02/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the assessment, every trainee should score the Recommended Pass % aggregate for the Qualification.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification.

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Minimum Aggregate Passing % at QP Level : 70

(**Please note:** Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SGJ/N1209.Introduce Solar Energy applications and importance of site survey	30	20	0	0	50	11
SGJ/N1210.Introduce Solar Photovoltaic Plant	25	25	0	0	50	11
SGJ/N1211.Assess shadow free area at solar project site	35	15	0	0	50	11
SGJ/N1212.Perform the use of tools for site survey	25	25	0	0	50	11
SGJ/N1213.Perform load analysis for small solar consumers	25	25	0	0	50	11
SGJ/N1214.Assess Consumers need for solar power	33	17	0	0	50	11
SGJ/N1215.Prepare a site survey report	35	15	0	0	50	11
SGJ/N1216.Maintain Personal Health & Safety while performing site survey at solar project site	27	23	0	0	50	11
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	12
Total	255	195	-	-	450	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.